

10 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION

KEY FINDINGS

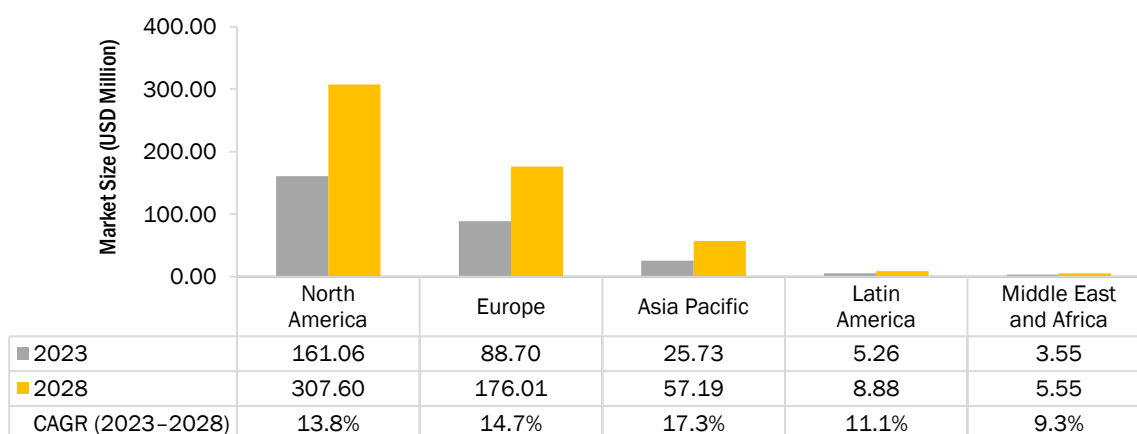
- North America accounted for the largest share of 56.9% of the microbiome sequencing services market in 2022. This market is projected to reach USD 307.60 million by 2028 from USD 161.06 million in 2023, at a CAGR of 13.8%.
- The established research infrastructure, technological advancements, and demand for precision medicine are driving the growth of the market in North America.
- Europe accounted for the second-largest share of 31.1% of the microbiome sequencing services market in 2022. This market is projected to reach USD 176.01 million by 2028, at a CAGR of 14.7% during the forecast period.
- The Asia Pacific market is estimated to grow at the highest CAGR of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine in the region.

10.1 INTRODUCTION

The global microbiome sequencing services market has been segmented into five major regional segments, namely, North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. In 2022, North America dominated the market with the largest share of 56.9%, followed by Europe (31.1%). Factors such as the established research infrastructure, technological advancements, demand for precision medicine, supportive regulatory environment, robust biotechnology and healthcare industry, and availability of funding and investment are driving the growth of the microbiome sequencing services market in North America.

The microbiome sequencing services market in the Asia Pacific is estimated to register the highest growth rate of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

FIGURE 25 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 87 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%
Europe	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%
Asia Pacific	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%
Latin America	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%
Middle East and Africa	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2 NORTH AMERICA

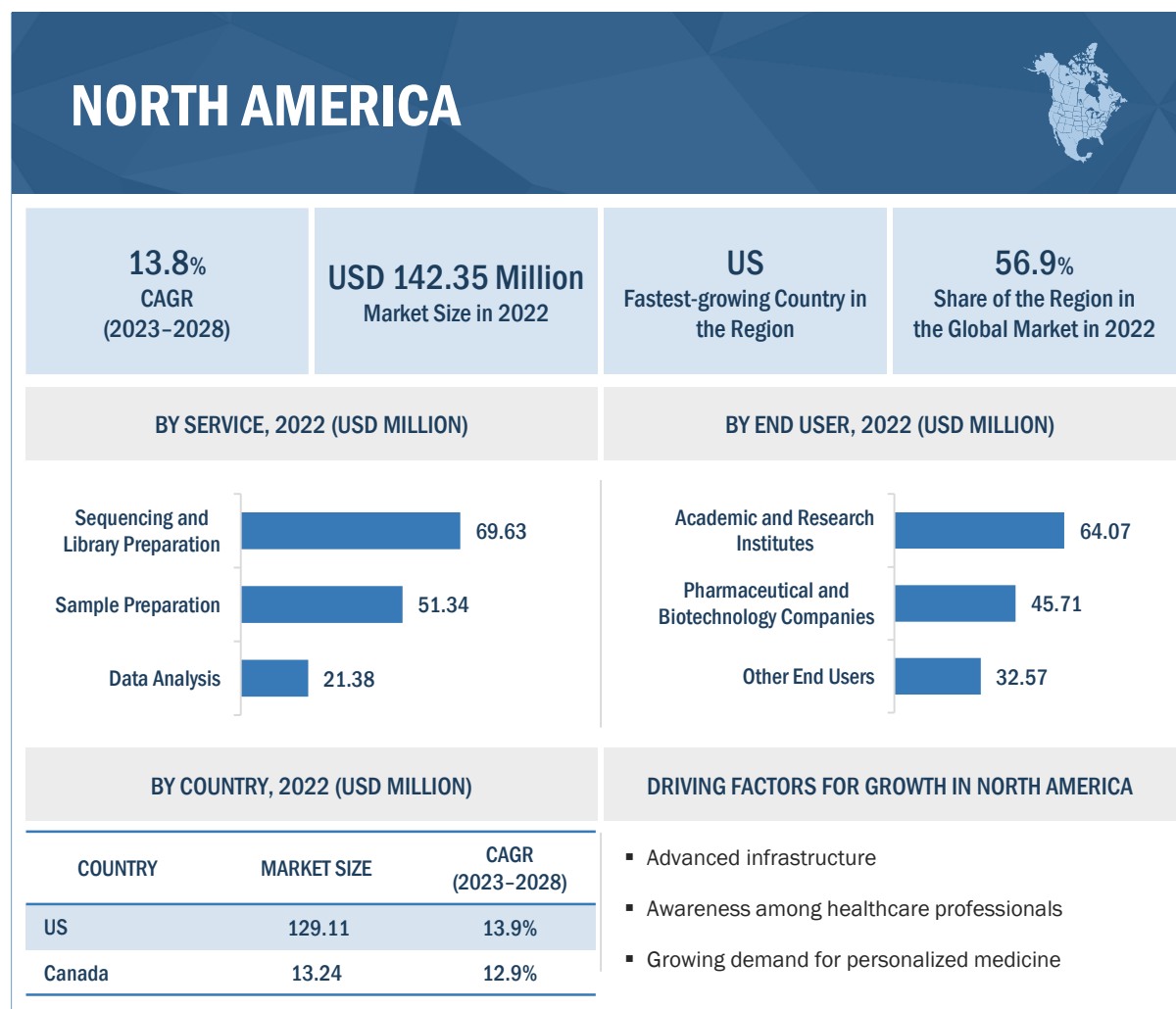
North America has emerged as a leading market for microbiome sequencing services, driven by a combination of factors. One of the key drivers is the well-established healthcare and biotechnology industry in the region, with advanced infrastructure, cutting-edge research institutions, and a strong ecosystem of biotechnology companies and CROs specializing in microbiome research. The presence of top academic institutions, research hospitals, and private companies in North America with dedicated microbiome research programs and expertise has contributed to the growth of the microbiome sequencing services market. In 2022, the National Library of Medicine released a research paper that discussed the correlation of gut microbiome diversity and abundance with Gray Matter Volume (GMV) in older adults with depression.

Another factor driving North America's leadership in this market is the high level of awareness among healthcare professionals and the general population about the importance of the human microbiome in health and disease. There has been a growing recognition of the role of the microbiome in various conditions, including gastrointestinal disorders, metabolic diseases, neurodegenerative diseases, and immune-related disorders. This increased awareness has led to a rising demand for microbiome sequencing services to better understand the complex interactions between the human microbiome and host health, as well as to develop targeted interventions and therapies.

Despite the positive drivers, there are also some restraints that can impact the microbiome sequencing services market in North America. One of the major challenges is the high cost associated with certain microbiome sequencing services, which can limit the adoption of these services for some customers, especially smaller research institutions or clinics with limited budgets. Regulatory and ethical considerations related to the use of human microbiome data, such as data privacy and consent issues, can also pose challenges in the market. However, there also are significant opportunities for growth in the North American microbiome sequencing services market. Advancements in sequencing technologies, data analysis, and bioinformatics are enabling more accurate and cost-effective microbiome sequencing, which can propel the adoption of these services. Increasing research collaborations between academia, industry, and government institutions in North America can also lead to novel discoveries and applications of microbiome sequencing, thus creating new market opportunities. The rising demand for personalized medicine approaches, where understanding an individual's microbiome can inform tailored treatment plans, presents a significant growth opportunity for microbiome sequencing services.

North America is home to several key companies that play a pivotal role in the microbiome sequencing services market. These companies, including Charles River Laboratories, CosmosID, and CD Genomics, are involved in the development and commercialization of microbiome sequencing technologies, platforms, and services. They are significantly contributing to the growth and innovation in the North American microbiome sequencing services market through their research, product offerings, and collaborations with academic and industry partners.

FIGURE 26 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Centre of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 88 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%
Canada	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%
Total	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 89 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	45.42	51.34	58.01	65.60	74.25	84.33	96.11	110.02	13.7%
Sequencing and Library Preparation	61.38	69.63	78.94	89.58	101.75	115.96	132.62	152.34	14.1%
Data Analysis	18.95	21.38	24.11	27.21	30.73	34.83	39.61	45.25	13.4%
Total	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 90 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%
Whole-genome Sequencing	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%
Shotgun Sequencing	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%
Other Types	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Total	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 91 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	57.03	64.63	73.20	82.98	94.15	107.20	122.48	140.55	13.9%
Sequencing by Ligation	16.07	18.15	20.49	23.15	26.18	29.71	33.83	38.69	13.6%
Nanopore Sequencing	6.47	7.47	8.62	9.95	11.49	13.30	15.46	18.03	15.9%
Other Technologies	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%
Total	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 92 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	40.30	45.71	51.82	58.80	66.77	76.09	87.02	99.94	14.0%
Academic and Research Institutes	56.62	64.07	72.45	82.01	92.90	105.62	120.48	138.04	13.8%
Other End Users	28.83	32.57	36.79	41.58	47.05	53.41	60.85	69.62	13.6%
Total	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.1 US

10.2.1.1 Increasing awareness of microbiome research to drive growth

The US microbiome sequencing services market is a rapidly growing and dynamic market that is driven by several factors. One of the key drivers for market growth in the US is the increasing awareness and understanding of the role of the microbiome in human health. Research has shown that microorganisms residing in and on the human body play a crucial role in various physiological processes, including digestion, immunity, and metabolism. The increasing focus on gaining a deep understanding of this has driven the demand for microbiome sequencing services to study the composition and function of the microbiome.

NGS technologies, such as Illumina's MiSeq and HiSeq platforms, have revolutionized the field of microbiome research by enabling the high-throughput and cost-effective sequencing of microbial communities. These technologies have greatly expanded the capability to study the complex and diverse microbial populations in the human microbiome. Several key players operating in the US market are adopting strategies to develop advanced products and provide efficient metagenomic NGS solutions to their customers. For instance, in 2021, CosmosID (US) formed a partnership with Locus Biosciences (US) to provide long-term support for Locus' clinical trial initiatives. CosmosID's infrastructure for providing access to robust, compliant, and high-resolution microbiome analysis results will allow Locus to add further

insights and capabilities to its precision therapeutics platform. Using shotgun metagenomics, CosmosID will help Locus understand the efficacy of its products through the highest resolution microbiome analysis, starting with its urinary tract infection clinical program.

One of the main challenges to the growth of the US market is the lack of standardized methods and protocols for microbiome sample collection, DNA extraction, library preparation, and data analysis. This can result in variability and inconsistency in results, making it challenging to compare and interpret data across different studies or laboratories.

TABLE 93 US: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	41.16	46.57	52.66	59.59	67.50	76.73	87.52	100.25	13.7%
Sequencing and Library Preparation	55.63	63.15	71.65	81.37	92.49	105.48	120.73	138.78	14.1%
Data Analysis	17.17	19.39	21.88	24.72	27.94	31.70	36.09	41.26	13.5%
Total	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 94 US: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	47.70	54.05	61.22	69.40	78.74	89.64	102.42	117.52	13.9%
Whole-genome Sequencing	24.40	27.56	31.12	35.17	39.79	45.16	51.43	58.83	13.6%
Shotgun Sequencing	30.64	34.90	39.74	45.29	51.65	59.11	67.88	78.29	14.5%
Other Types	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Total	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 95 US: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	51.68	58.62	66.45	75.39	85.60	97.54	111.54	128.09	14.0%
Sequencing by Ligation	14.56	16.46	18.60	21.03	23.79	27.02	30.79	35.23	13.6%
Nanopore Sequencing	5.87	6.78	7.82	9.03	10.43	12.09	14.05	16.40	16.0%
Other Technologies	41.85	47.25	53.33	60.24	68.11	77.27	87.96	100.57	13.5%
Total	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 96 US: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	36.53	41.46	47.03	53.40	60.69	69.20	79.19	91.01	14.1%
Academic and Research Institutes	51.31	58.11	65.77	74.50	84.47	96.10	109.72	125.81	13.9%
Other End Users	26.12	29.54	33.39	37.78	42.78	48.61	55.42	63.47	13.7%
Total	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.2 CANADA

10.2.2.1 Increasing prevalence of chronic diseases to boost microbiome research

The microbiome sequencing services market in Canada has been witnessing significant growth in recent years, driven by various factors. One of the key drivers is the increasing prevalence of chronic diseases, such as obesity, diabetes, and inflammatory bowel disease (IBD), in Canada. As the composition of the gut microbiome is linked to these diseases, microbiome sequencing can provide valuable insights into the underlying mechanisms and potential therapeutic strategies.

Advancements in sequencing technologies, including SBS, have made the high-throughput, cost-effective sequencing of microbial communities possible. The availability of NGS platforms has enabled the large-scale sequencing of the microbiome. The Government of Canada has undertaken several initiatives to improve the healthcare system in the country. For instance, in 2021, the government initiated the third phase of the Strategic Plan 2021–2026. This plan sets out priorities that align with the CIHR Strategic Plan 2021–2031 and the STBBI Research Initiative Strategic Plan. The plan aims to achieve excellence in infection and immunity research. In January 2019, the government announced an investment of USD 18 million over five years in microbiome research. This investment supported seven research teams from

across the country that studied the microbiome to better understand its contribution to childhood asthma, cervical cancer, diabetes, youth inflammatory bowel disease, and maternal malnutrition. The investment also helped establish a Microbiome Research Core at the University of Calgary and support the teams by providing expertise in study design, sample collection, and analysis, and, eventually, a large repository of microbes.

The microbiome sequencing services market in Canada faces certain restraints, including challenges in standardization, data analysis, and interpretation. Additionally, regulatory considerations and ethical concerns related to the use of human microbiome data can impact market growth.

TABLE 97 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	4.25	4.78	5.35	6.01	6.75	7.60	8.60	9.76	12.8%
Sequencing and Library Preparation	5.75	6.48	7.29	8.21	9.26	10.48	11.89	13.56	13.2%
Data Analysis	1.78	1.99	2.22	2.49	2.78	3.13	3.53	3.99	12.4%
Total	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 98 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	4.93	5.54	6.22	7.00	7.88	8.90	10.08	11.48	13.0%
Whole-genome Sequencing	2.53	2.83	3.16	3.54	3.97	4.46	5.03	5.70	12.5%
Shotgun Sequencing	3.17	3.58	4.05	4.57	5.18	5.88	6.70	7.67	13.6%
Other Types	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 99 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	5.34	6.01	6.75	7.59	8.55	9.66	10.94	12.46	13.0%
Sequencing by Ligation	1.51	1.69	1.89	2.13	2.39	2.69	3.04	3.45	12.8%
Nanopore Sequencing	0.61	0.70	0.80	0.92	1.06	1.22	1.41	1.63	15.4%
Other Technologies	4.33	4.85	5.42	6.07	6.80	7.65	8.62	9.77	12.5%
Total	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 100 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	3.77	4.25	4.79	5.39	6.09	6.89	7.83	8.93	13.3%
Academic and Research Institutes	5.31	5.96	6.69	7.51	8.44	9.51	10.76	12.23	12.8%
Other End Users	2.70	3.03	3.39	3.80	4.27	4.80	5.43	6.15	12.6%
Total	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.3 NORTH AMERICA: IMPACT OF RECESSION

North America is very likely to enter a recession in mid-2023, with higher-than-expected inflation, especially in the US, triggering a tightening of global financial conditions. According to the International Monetary Fund (IMF) World Economic Outlook 2022, in the US, reduced household purchasing power and tighter monetary policy will drive GDP growth down from 2.3% in 2022 to 1% in 2023. Similarly, in Canada, growth in real GDP is projected to slow from 3.2% in 2022 to 1.0% in 2023 before strengthening to 1.3% in 2024. However, people's spending on healthcare is less likely to be affected by macroeconomic headwinds as diseases will always exist, and treatments and medicines will always be in demand, irrespective of economic turbulence. Health expenditure in North America could exceed economic growth by up to 5.9% in 2023. National health expenditure could grow at a rate of 7.1% over the next five years, from 2022 to 2027, compared with an expected economic growth rate of 4.7%.

The US government is taking action to combat the upcoming recession. In August 2022, the US government passed the Inflation Reduction Act, which features significant changes to federal prescription drug pricing policies that can reduce costs for individuals receiving care through Medicare. The bill also includes a three-year extension of expanded Affordable Care Act (ACA) healthcare benefits through 2025. Additionally, more mergers and acquisitions are expected in 2023 in the US, with a deal value of USD 225 billion to USD 275 billion across all subsectors. Companies such as Johnson & Johnson and Pfizer have entered deals with Abiomed and Biohaven Pharmaceuticals for USD 18.9 billion and USD 12.2 billion,

respectively. In the pharma and life science sector, the US market is projected to exhibit higher growth in 2023. This is attributed to pharma and biotech M&A growth due to an increased focus on oncology and immunology.

10.3 EUROPE

The microbiome sequencing services market in Europe has been growing steadily in recent years, driven by several factors, such as the increasing awareness of and interest in personalized health and wellness, the growing adoption of microbiome-based therapies, and advancements in sequencing technologies and bioinformatics analysis. Consumers in Europe are becoming more aware of the role of the microbiome in health and disease and are seeking personalized solutions to optimize their microbiome composition and function. This has led to the growth of direct-to-consumer microbiome sequencing services, which offer comprehensive analysis and personalized recommendations based on an individual's microbiome composition.

Due to the role of the microbiome in various diseases and conditions, such as IBD, obesity, and autism, the focus on developing microbiome-based therapies to treat these conditions has increased in Europe in recent years. This has led to an increase in the demand for microbiome sequencing services in R&D and clinical applications.

Major companies are playing a significant role in driving innovation in this market and offering advanced technologies and services across Europe. In April 2022, Microba Life Sciences and Ginkgo Bioworks announced a partnership to identify single-strain, live bacteria product (LBP) candidates against autoimmune diseases. The collaboration aimed to build on Microba's precision approach to LBP development with an in-depth evaluation of the company's strains using Ginkgo's high-throughput and automated screening capabilities. Through this partnership, Ginkgo provided high-throughput screening for Microba's proprietary library of human microbiome-isolated strains with the goal of improving treatment for autoimmune diseases such as lupus, psoriatic arthritis, and certain autoimmune liver diseases. Microba leveraged Ginkgo's high-throughput anaerobic culturing, multi-omics data collection and analysis, functional bioassay screening, and media and fermentation optimization capabilities to generate data sets that may help characterize potential therapeutic and non-therapeutic uses of the strains. Moreover, in 2020, Microba entered into an agreement with SYNLAB Diagnostics Globales to deliver Microba's testing services to healthcare providers in Spain, with an option to formalize distribution agreements across their affiliate organizations, which operate in 36 countries.

The use of microbiome sequencing in clinical applications is subject to regulatory oversight, and there are concerns about the accuracy and reliability of microbiome analysis and interpretation. Additionally, the collection and analysis of personal microbiome data raise concerns about data privacy and security. These factors are expected to restrain market growth in Europe.

TABLE 101 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%
UK	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%
France	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%
Italy	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%
Spain	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%
Rest of Europe	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 102 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	24.57	27.99	31.87	36.32	41.42	47.40	54.44	62.80	14.5%
Sequencing and Library Preparation	33.26	38.09	43.60	49.94	57.25	65.87	76.04	88.17	15.1%
Data Analysis	10.37	11.72	13.24	14.97	16.93	19.22	21.89	25.05	13.6%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 103 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%
Whole-genome Sequencing	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%
Shotgun Sequencing	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%
Other Types	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 104 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	30.83	35.22	40.20	45.94	52.54	60.30	69.44	80.32	14.8%
Sequencing by Ligation	8.75	9.96	11.32	12.89	14.69	16.80	19.27	22.21	14.4%
Nanopore Sequencing	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%
Other Technologies	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 105 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	21.68	24.91	28.62	32.90	37.85	43.70	50.62	58.90	15.5%
Academic and Research Institutes	30.81	35.05	39.86	45.38	51.70	59.10	67.81	78.13	14.4%
Other End Users	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%
Total	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.1 GERMANY

10.3.1.1 Rapidly increasing government funding to support growth

Germany is the largest contributor to the European microbiome sequencing services market, with an estimated share of 21.8% in 2023. The microbiome sequencing services market in Germany is estimated to grow from USD 19.31 million in 2023 to USD 38.88 million by 2028, at a CAGR of 15.0%.

The microbiome sequencing services market in Germany is growing rapidly, driven by the increasing demand for microbiome analysis in research and clinical applications. The German government is making significant investments in the development of innovative technologies and research infrastructure, including microbiome research. This funding is driving the growth of the microbiome sequencing services market, as it allows researchers and companies to access the resources they need to advance in the field. Also, advances in sequencing technologies and data analysis tools are making microbiome analysis faster, more accurate, and more accessible.

The high cost of sequencing technologies and the lack of standardization in data analysis and interpretation are expected to restrain the growth of this market. Despite these challenges, there are significant opportunities for growth in the German microbiome sequencing services market, including the

development of new technologies and applications, as well as the increasing collaborations between researchers and clinicians.

TABLE 106 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	5.32	6.08	6.95	7.94	9.08	10.42	12.01	13.89	14.9%
Sequencing and Library Preparation	7.20	8.26	9.47	10.87	12.49	14.40	16.66	19.35	15.4%
Data Analysis	2.23	2.54	2.89	3.28	3.74	4.27	4.90	5.64	14.3%
Total	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 107 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	6.16	7.06	8.08	9.26	10.62	12.22	14.11	16.37	15.2%
Whole-genome Sequencing	3.17	3.61	4.11	4.68	5.33	6.10	7.00	8.08	14.5%
Shotgun Sequencing	3.96	4.57	5.26	6.06	6.99	8.09	9.40	10.96	15.8%
Other Types	1.46	1.65	1.86	2.10	2.37	2.68	3.04	3.47	13.3%
Total	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 108 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	6.68	7.65	8.77	10.04	11.52	13.26	15.32	17.77	15.2%
Sequencing by Ligation	1.89	2.16	2.46	2.80	3.20	3.67	4.22	4.87	14.7%
Nanopore Sequencing	0.76	0.89	1.04	1.23	1.44	1.70	2.00	2.38	17.9%
Other Technologies	5.43	6.18	7.04	8.02	9.15	10.47	12.02	13.86	14.5%
Total	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 109 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	4.71	5.41	6.22	7.15	8.22	9.49	11.00	12.80	15.5%
Academic and Research Institutes	6.65	7.60	8.68	9.92	11.35	13.03	15.01	17.37	14.9%
Other End Users	3.39	3.87	4.41	5.03	5.74	6.57	7.55	8.72	14.6%
Total	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.2 UK

10.3.2.1 Significant government investments in genomics research to propel growth

In 2022, the UK accounted for a share of 19.5% of the microbiome sequencing services market in Europe. This market is projected to reach USD 35.91 million by 2028 from USD 17.43 million in 2023, at a CAGR of 15.6% during the forecast period.

The UK government's significant investment in genomics research has helped fund and support several innovative microbiome sequencing companies and research initiatives. In addition, the country's strong academic and research base, combined with its robust healthcare system, has created a supportive environment for microbiome sequencing research and development.

The increasing prevalence of chronic diseases in the UK, such as obesity, diabetes, and cardiovascular disease, has also driven the demand for microbiome analysis to better understand these diseases and develop new treatments. In March 2022, Genetic Signatures (Australia) joined BioHub Birmingham as a launchpad for a planned further expansion into Europe, the Middle East, and Africa. BioHub is the first physical footprint within the UK and Europe for Genetic Signatures, which already provides the EasyScreen

range of testing kits to countries in Europe and provides clinical diagnostics to the NHS for respiratory and gastro-intestinal diseases, as well as sexually transmitted infections.

The high cost of sequencing technologies and the lack of standardization in data analysis and interpretation are expected to restrain market growth in the UK.

TABLE 110 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	4.76	5.46	6.27	7.20	8.27	9.54	11.03	12.82	15.4%
Sequencing and Library Preparation	6.44	7.43	8.56	9.87	11.40	13.20	15.35	17.92	15.9%
Data Analysis	2.00	2.28	2.60	2.97	3.39	3.89	4.47	5.17	14.7%
Total	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 111 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	5.50	6.34	7.29	8.40	9.68	11.20	13.00	15.15	15.7%
Whole-genome Sequencing	2.84	3.24	3.71	4.24	4.85	5.57	6.41	7.42	14.9%
Shotgun Sequencing	3.54	4.11	4.75	5.51	6.39	7.43	8.67	10.17	16.4%
Other Types	1.31	1.48	1.68	1.90	2.14	2.43	2.77	3.17	13.6%
Total	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 112 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	5.97	6.87	7.91	9.10	10.48	12.12	14.06	16.38	15.7%
Sequencing by Ligation	1.69	1.94	2.22	2.55	2.93	3.37	3.90	4.52	15.3%
Nanopore Sequencing	0.68	0.80	0.95	1.12	1.33	1.58	1.88	2.24	18.8%
Other Technologies	4.86	5.56	6.35	7.27	8.32	9.56	11.03	12.76	15.0%
Total	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 113 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	4.21	4.86	5.62	6.50	7.52	8.73	10.17	11.90	16.2%
Academic and Research Institutes	5.95	6.83	7.84	8.99	10.33	11.91	13.77	16.00	15.3%
Other End Users	3.04	3.48	3.98	4.55	5.22	6.00	6.92	8.01	15.0%
Total	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.3 FRANCE

10.3.3.1 Growing number of microbiome sequencing startups to propel growth

The market in France is projected to reach USD 27.04 million by 2028 from USD 13.58 million in 2023, at a CAGR of 14.8% during the forecast period.

The presence of a strong healthcare system and the increasing demand for personalized medicine are the key factors driving market growth in France. Several companies in France are focusing on developing new products and services that use microbiome data to provide personalized diet and lifestyle recommendations to individuals based on their unique microbiome makeup. Microbiome sequencing startups and spin-offs in France are developing innovative technologies and services for microbiome research and analysis and creating new opportunities for market growth.

In September 2022, the Le French Gut project was launched by the *Institut national de la recherche agronomique* (INRAE). This project will contribute to the international Million Microbiome of Humans Project. INRAE's research unit, MetaGenoPolis, has initiated the project in association with AP-HP. The Le French Gut project aims to better understand healthy gut microbiota and model and predict changes in gut microbiota associated with diseases.

The lack of funding and support for research and development in this area and a lack of standardization in data analysis and interpretation are the key factors restraining market growth in France.

TABLE 114 HUMAN MICROBIOME STARTUPS IN FRANCE

COMPANY NAME	THERAPEUTIC AREA
Enterome	Develops microbiome-based diagnostics and therapeutics for chronic diseases
Eligo Bioscience	Develops microbiome-based therapies using CRISPR-Cas technology
BiomX	Develops microbiome-based therapeutics for bacterial infections and inflammatory bowel disease
TargEDys	Develops microbiome-based therapies for obesity and related disorders
LNC Therapeutics	Develops microbiome-based therapies for metabolic and inflammatory diseases
YSOPIA Bioscience	Develops microbiome-based therapies for cancer and immune disorders
Pherecydes Pharma	Develops microbiome-based phage therapies for bacterial infections
Predilife	Provides microbiome-based genetic testing and personalized health advice
InnaVirVax	Develops microbiome-based immunotherapies for chronic viral infections
TreeFrog Therapeutics	Develops microbiome-based therapies for cell therapy and regenerative medicine

TABLE 115 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	3.76	4.28	4.88	5.57	6.35	7.27	8.36	9.65	14.6%
Sequencing and Library Preparation	5.09	5.83	6.67	7.65	8.77	10.10	11.66	13.53	15.2%
Data Analysis	1.58	1.79	2.03	2.29	2.60	2.96	3.37	3.86	13.7%
Total	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 116 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	4.34	4.97	5.68	6.50	7.45	8.56	9.87	11.43	15.0%
Whole-genome Sequencing	2.25	2.55	2.89	3.27	3.72	4.23	4.84	5.56	14.0%
Shotgun Sequencing	2.80	3.22	3.71	4.27	4.92	5.69	6.60	7.69	15.7%
Other Types	1.04	1.17	1.31	1.46	1.64	1.84	2.08	2.35	12.5%
Total	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 117 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	4.72	5.39	6.15	7.03	8.05	9.24	10.64	12.31	14.9%
Sequencing by Ligation	1.34	1.52	1.73	1.98	2.26	2.58	2.97	3.43	14.6%
Nanopore Sequencing	0.53	0.63	0.74	0.88	1.03	1.22	1.45	1.72	18.3%
Other Technologies	3.84	4.36	4.95	5.62	6.39	7.28	8.34	9.58	14.1%
Total	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 118 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	3.32	3.81	4.38	5.04	5.79	6.69	7.75	9.02	15.5%
Academic and Research Institutes	4.71	5.36	6.10	6.95	7.93	9.07	10.42	12.02	14.5%
Other End Users	2.40	2.73	3.10	3.52	4.00	4.56	5.22	6.00	14.2%
Total	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.4 ITALY

10.3.4.1 Increasing use of microbiome services in clinical research to drive growth

The market in Italy is projected to reach USD 7.46 million by 2028 from USD 3.81 million in 2023, at a CAGR of 14.4% during the forecast period.

The increasing use of microbiome services in clinical research is a key factor driving market growth in Italy. Italian researchers are using microbiome analysis to gain new insights into the role of the microbiome in a range of diseases, from IBD to cancer. This is driving the demand for high-quality microbiome sequencing services from both academic and commercial research institutions across the country. In 2022, the National Center for Biotechnology Information published a research article that explained the microbiota-gut axis in premature infants and its physio-pathological implications. In 2021, research on gut microbial dysbiosis in the pathogenesis of gastrointestinal dysmotility and metabolic disorders was conducted. It explained how constant communication between the gut microbiota-derived metabolites and human body systems regulates the physiological aspects of health and disease.

Italian companies are at the forefront of developing new microbiome-based therapies, particularly for IBD. This is driving the demand for high-quality microbiome sequencing services and creating new opportunities for collaborations and partnerships between Italian biotechnology companies and international microbiome sequencing providers.

The high cost associated with sequencing services is limiting the market growth. Although the cost of sequencing has decreased over the years, it is still relatively high and can be a significant barrier for some customers.

TABLE 119 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.20	1.37	1.55	1.77	2.02	2.31	2.66	14.3%
Sequencing and Library Preparation	1.43	1.64	1.87	2.14	2.44	2.80	3.23	3.73	14.8%
Data Analysis	0.45	0.50	0.57	0.64	0.72	0.82	0.93	1.07	13.4%
Total	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 120 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.40	1.59	1.82	2.08	2.38	2.74	3.16	14.7%
Whole-genome Sequencing	0.63	0.72	0.81	0.91	1.03	1.17	1.33	1.53	13.5%
Shotgun Sequencing	0.79	0.91	1.04	1.19	1.37	1.59	1.83	2.13	15.4%
Other Types	0.29	0.33	0.37	0.41	0.45	0.51	0.57	0.64	11.9%
Total	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 121 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.33	1.51	1.72	1.96	2.24	2.56	2.94	3.39	14.5%
Sequencing by Ligation	0.38	0.43	0.49	0.55	0.63	0.72	0.83	0.95	14.3%
Nanopore Sequencing	0.15	0.18	0.21	0.25	0.29	0.34	0.40	0.48	18.1%
Other Technologies	1.08	1.23	1.39	1.57	1.78	2.02	2.30	2.64	13.7%
Total	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 122 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.93	1.07	1.23	1.41	1.62	1.87	2.16	2.51	15.3%
Academic and Research Institutes	1.33	1.51	1.71	1.94	2.21	2.52	2.88	3.31	14.1%
Other End Users	0.68	0.77	0.87	0.98	1.11	1.26	1.44	1.65	13.7%
Total	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.5 SPAIN

10.3.5.1 Increasing prevalence of chronic diseases to drive market

The market in Spain is projected to reach USD 5.65 million by 2028 from USD 2.91 million in 2023, at a CAGR of 14.2% during the forecast period.

The increasing prevalence of chronic diseases, growing demand for personalized medicine, and rising awareness about the role of gut microbiomes in human health are the key factors driving market growth in Spain. The prevalence of chronic diseases, such as obesity, diabetes, and IBD, is increasing in Spain. These conditions are associated with dysbiosis of the gut microbiome, and understanding the composition and function of the microbiome can provide valuable insights into their pathogenesis and potential treatments. However, a lack of awareness about microbiome sequencing services among the general population and healthcare professionals is expected to limit the demand for these services.

TABLE 123 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.81	0.92	1.05	1.19	1.35	1.53	1.75	2.01	14.0%
Sequencing and Library Preparation	1.10	1.26	1.43	1.64	1.87	2.14	2.46	2.84	14.7%
Data Analysis	0.34	0.39	0.43	0.49	0.55	0.62	0.70	0.79	12.8%
Total	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 124 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.94	1.07	1.22	1.39	1.58	1.81	2.08	2.40	14.5%
Whole-genome Sequencing	0.49	0.55	0.62	0.70	0.79	0.89	1.01	1.15	13.2%
Shotgun Sequencing	0.61	0.70	0.80	0.91	1.05	1.21	1.40	1.62	15.3%
Other Types	0.23	0.25	0.28	0.31	0.34	0.38	0.43	0.48	11.4%
Total	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 125 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.02	1.16	1.32	1.50	1.71	1.95	2.24	2.58	14.3%
Sequencing by Ligation	0.29	0.33	0.37	0.42	0.48	0.55	0.62	0.72	14.0%
Nanopore Sequencing	0.12	0.14	0.16	0.19	0.22	0.26	0.31	0.37	18.0%
Other Technologies	0.83	0.94	1.06	1.20	1.35	1.53	1.74	1.99	13.4%
Total	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 126 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.72	0.82	0.94	1.08	1.24	1.42	1.65	1.91	15.2%
Academic and Research Institutes	1.02	1.16	1.31	1.48	1.68	1.91	2.18	2.50	13.8%
Other End Users	0.52	0.59	0.66	0.75	0.85	0.96	1.09	1.24	13.3%
Total	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.6 REST OF EUROPE

The Rest of Europe region comprises Switzerland, Sweden, Poland, Austria, Belgium, Denmark, and Russia, among other European countries. Market growth in this region is primarily driven by an increase in NGS-based research, an improving clinical reimbursement structure, and government investments to support genomics research. In May 2019, Novigenix announced plans to develop a new version of Colox, its mRNA-based assay for early colorectal cancer detection that applies next-generation sequencing. The company partnered with six European collaborators, including the Swiss Federal Institute of Technology, the Lausanne University Hospital, and the Swiss Institute of Bioinformatics, to recruit 1,600 individuals for CRC screening as part of the validation study. In January 2019, the Novo Nordisk Foundation approved a framework grant of USD 157.8 million over 4.5 years to establish and operate the infrastructure of the National Genome Centre. Such developments are expected to drive market growth in this region.

TABLE 127 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	8.86	10.04	11.36	12.87	14.60	16.62	18.97	21.76	13.9%
Sequencing and Library Preparation	12.00	13.68	15.59	17.77	20.28	23.23	26.69	30.80	14.6%
Data Analysis	3.76	4.21	4.72	5.29	5.93	6.66	7.52	8.52	12.5%
Total	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 128 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	10.22	11.64	13.24	15.08	17.18	19.65	22.55	25.99	14.4%
Whole-genome Sequencing	5.32	5.98	6.72	7.55	8.50	9.59	10.86	12.35	12.9%
Shotgun Sequencing	6.60	7.57	8.67	9.95	11.42	13.14	15.18	17.62	15.2%
Other Types	2.48	2.74	3.03	3.36	3.71	4.12	4.58	5.12	11.0%
Total	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 129 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	11.11	12.62	14.34	16.30	18.54	21.16	24.24	27.89	14.2%
Sequencing by Ligation	3.16	3.58	4.05	4.59	5.19	5.91	6.74	7.72	13.8%
Nanopore Sequencing	1.26	1.49	1.75	2.06	2.43	2.86	3.39	4.02	18.1%
Other Technologies	9.09	10.24	11.53	12.99	14.65	16.57	18.81	21.44	13.2%
Total	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 130 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	7.79	8.93	10.23	11.73	13.46	15.50	17.91	20.77	15.2%
Academic and Research Institutes	11.14	12.59	14.22	16.08	18.20	20.66	23.55	26.94	13.6%
Other End Users	5.70	6.41	7.21	8.12	9.15	10.34	11.73	13.36	13.1%
Total	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.7 EUROPE: IMPACT OF RECESSION

Europe was expected to witness an economic recession in the last quarter of 2022, according to the European Commission's autumn economic forecast. In Europe, inflation is expected to have a higher impact on all the major economies, such as the UK, Germany, France, Italy, and Spain. The region is expected to experience a recession till mid-2024.

Healthcare and pharmaceutical spending in European countries has been increasing consistently. The coming recession is expected to impact the healthcare and pharmaceutical industry with reduced health budgets, workers' wages, healthcare provision, pharmaceutical spending, and increased hours worked by healthcare professionals. Thus, there will be a slight impact on the healthcare market, including on sequencing technologies, with a slow market growth rate in 2023. However, the market growth rate is expected to recover in 2025-26.

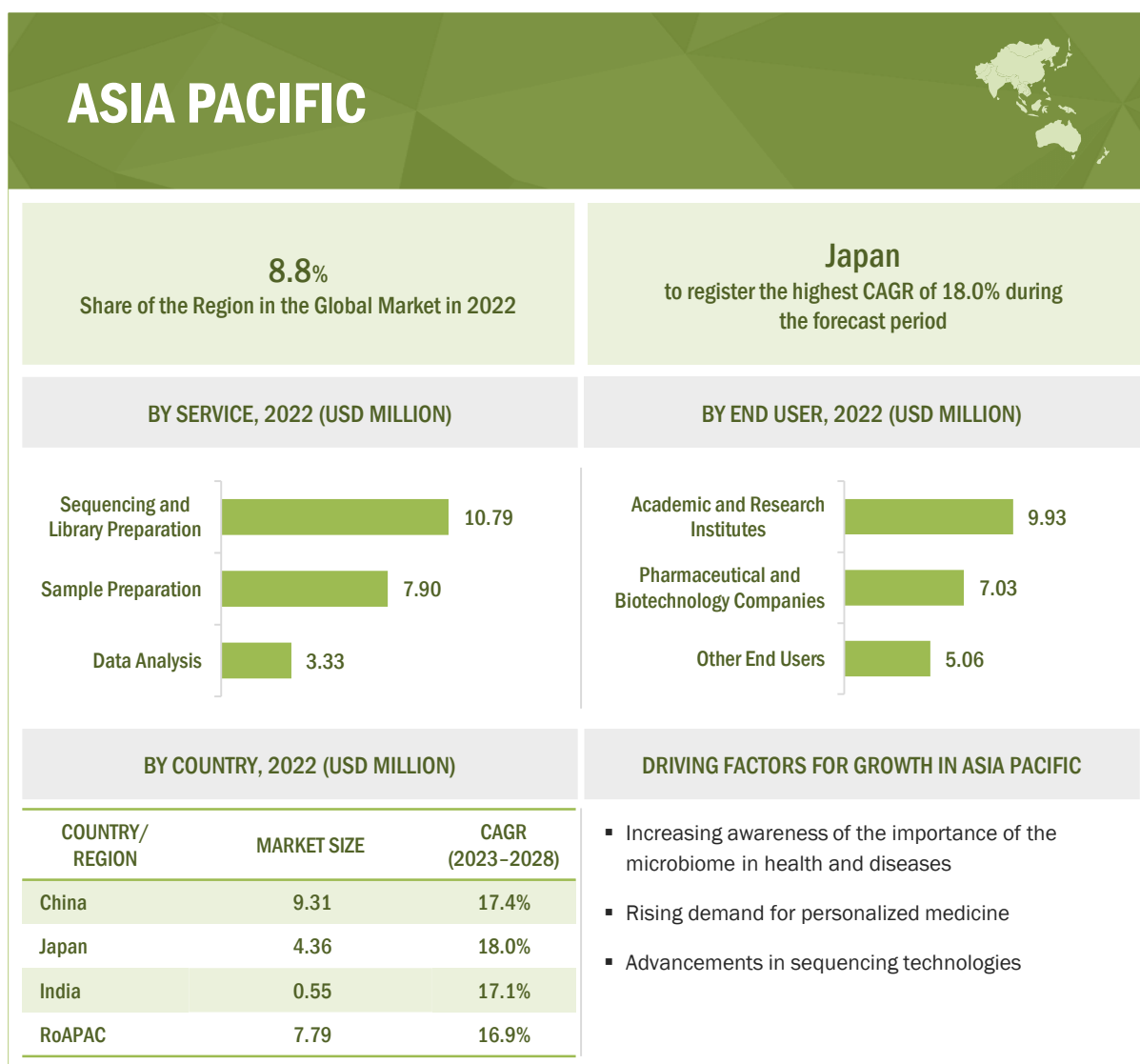
This impact will be balanced by other factors, such as the expansion of business operations in Europe, an increase in clinical trials, and growing drug pipelines. International pharmaceutical companies are working toward expanding their operations in Europe. For instance, in December 2022, Pfizer announced an investment of USD 1.2 billion for its manufacturing site in Puurs, Belgium. This expansion will occur over the next three years, with some projects already started while others are scheduled to begin in early 2023. Such developments will help European healthcare markets to withstand the effects of an economic downturn.

10.4 ASIA PACIFIC

The Asia Pacific microbiome sequencing services market is estimated to reach USD 57.19 million by 2028 from USD 25.73 million in 2023, at a CAGR of 17.3% during the forecast period. In 2022, China accounted for the largest share of 42.3% of the microbiome sequencing services market in the Asia Pacific, followed by Japan with a share of 19.8%.

The Asia Pacific region is expected to be the fastest-growing market for microbiome sequencing services due to the increasing awareness of the importance of the microbiome in health and disease, rising demand for personalized medicine, and advancements in sequencing technologies. Another key driver for the Asia Pacific microbiome sequencing services market is the region's large and rapidly aging population. As people age, they become more susceptible to a range of diseases and conditions that may be influenced by the microbiome. This has led to an increased focus on preventative healthcare and personalized medicine, which is driving the demand for microbiome sequencing services. The development of high-throughput sequencing technologies has made it possible to sequence the entire microbiome in a single sample, enabling researchers to more accurately and comprehensively characterize the microbial communities that are present in samples.

Challenges in terms of data protection and privacy, as well as the high cost of microbiome sequencing services, are expected to restrain the growth of the Asia Pacific market. In addition, there may be cultural and linguistic barriers to adoption in certain regions, as well as concerns about the accuracy and clinical utility of microbiome sequencing data.

FIGURE 27 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 131 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%
Japan	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%
India	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%
Rest of Asia Pacific	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 132 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%
Sequencing and Library Preparation	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%
Data Analysis	2.89	3.33	3.83	4.40	5.06	5.84	6.75	7.83	15.4%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 133 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Whole-genome Sequencing	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Shotgun Sequencing	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Other Types	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 134 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	8.47	9.93	11.63	13.62	15.96	18.75	22.09	26.12	17.6%
Sequencing by Ligation	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%
Nanopore Sequencing	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%
Other Technologies	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 135 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	5.92	7.03	8.33	9.86	11.68	13.87	16.52	19.74	18.8%
Academic and Research Institutes	8.52	9.93	11.55	13.44	15.64	18.25	21.36	25.09	16.8%
Other End Users	4.37	5.06	5.85	6.77	7.84	9.10	10.59	12.36	16.1%
Total	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.1 CHINA

10.4.1.1 Increasing investments in R&D to drive growth

The microbiome sequencing services market in China has seen significant growth in recent years. The rapidly expanding healthcare industry in China has led to increased investment in the research and development of new microbiome sequencing technologies. China is home to one of the world's fastest-growing biotechnology markets. In 2020, Chinese genetic testing companies in the primary market raised over RMB 19.3 billion (USD 3 billion) in financing—an annual growth rate of 153% compared to 2019.

The rising incidence of chronic diseases in China, such as cancer and diabetes, has fueled the demand for more accurate and efficient diagnostic tools. Additionally, the Chinese government has placed a strong emphasis on promoting the development of the biotechnology and life science industries, which has created opportunities for companies in the microbiome sequencing services market to receive funding and support for R&D efforts. This has also encouraged collaborations between academic institutions and private companies in China, which has facilitated the development of new technologies and therapies. For instance, Biomica Ltd. announced the signing of a definitive agreement for a USD 20 million financing round to be led by Shanghai Healthcare Capital (SHC). The financing is subject to customary closing conditions, including clearance by Chinese regulatory authorities. The financing round will enable Biomica to forge ahead, developing its pipeline of microbiome-based therapeutics. Moreover, Siolta Therapeutics

announced that it raised USD 30 million in a series B investment round to support the clinical development of its lead product, STMC-103H. The series B funding involved participation from top-tier investors and venture firms, including series A investors Khosla Ventures and Marc Benioff, as well as new investors such as Seventure (Health for Life Capital Fund), SymBiosis, and Global Brain (Kirin Health Innovation Fund/GB-VII). Siolta's live biotherapeutic product (LBP) platform is designed to develop microbiome-based medicines and diagnostics for the prevention and treatment of chronic diseases in targeted populations.

One of the main restraints for the microbiome sequencing services market in China is the lack of regulatory oversight and standardization in the industry. This has created uncertainty for investors and may hinder the growth of the market in the long term. Additionally, the high cost of technology and limited reimbursement options may limit the accessibility of microbiome sequencing services among patients.

Despite these challenges, there still are several opportunities for the growth of the microbiome sequencing services market in China. For instance, the growing demand for personalized medicine and precision healthcare has increased the focus on individualized microbiome analysis, which has the potential to drive the demand for microbiome sequencing services. The rapid growth of the Chinese middle class has led to an increased demand for high-quality healthcare services, which could further accelerate the growth of the microbiome sequencing services market.

TABLE 136 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Sequencing and Library Preparation	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Data Analysis	1.22	1.41	1.62	1.87	2.15	2.49	2.89	3.36	15.7%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 137 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Whole-genome Sequencing	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Shotgun Sequencing	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Other Types	0.80	0.92	1.04	1.19	1.35	1.53	1.75	2.00	14.0%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 138 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.58	4.21	4.93	5.77	6.77	7.96	9.38	11.09	17.6%
Sequencing by Ligation	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Nanopore Sequencing	0.41	0.50	0.61	0.74	0.90	1.09	1.33	1.63	21.9%
Other Technologies	2.94	3.42	3.96	4.60	5.34	6.21	7.25	8.49	16.4%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 139 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.51	2.98	3.52	4.16	4.93	5.84	6.95	8.30	18.7%
Academic and Research Institutes	3.60	4.20	4.89	5.69	6.63	7.75	9.07	10.67	16.9%
Other End Users	1.84	2.14	2.48	2.87	3.33	3.87	4.51	5.28	16.3%
Total	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.2 JAPAN

10.4.2.1 Significant growth in geriatric population to present opportunities for market growth

The microbiome sequencing services market in Japan has been growing rapidly in recent years, driven by several factors. One key driver is the increasing prevalence of chronic diseases, such as cancer, diabetes, and IBD, which have been linked to dysbiosis or imbalances in the gut microbiome. This has led to a growing interest in the potential of microbiome analysis for personalized medicine and disease diagnosis and treatment. Another driver is the development of microbiome-based therapeutics, such as fecal microbiota transplantation (FMT), which have been approved in Japan for the treatment of recurrent *Clostridioides difficile* infections.

In addition to these drivers, Japan's aging population presents a unique opportunity for microbiome sequencing services. Japan's population is aging and shrinking fast. With a median age of 48.4 years, Japan's population is the oldest in the world. The government of Japan projects that there will be almost one elderly person for each person of working age by 2060. Over 29.18% of the total population is 65 years or older. As the population ages, the incidence of age-related diseases such as dementia and Parkinson's disease is increasing, and there is a growing interest in the potential of the gut microbiome to

play a role in these conditions. Japan's highly developed healthcare system and advanced research infrastructure also make it an attractive market for microbiome sequencing services.

However, there are several restraints to the growth of the microbiome sequencing services market in Japan. One is the lack of public awareness and understanding of the microbiome and its potential applications in healthcare. This has led to the slow adoption of microbiome analysis services by both patients and healthcare providers. In addition, regulatory hurdles and a lack of reimbursement for microbiome analysis services have hindered the growth of this market.

Despite these challenges, the increasing availability of NGS technologies and bioinformatics tools, which have greatly improved the speed and accuracy of microbiome analysis, are expected to offer potential growth opportunities for market growth in Japan.

TABLE 140 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
Sequencing and Library Preparation	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
Data Analysis	0.57	0.66	0.76	0.88	1.02	1.18	1.38	1.61	16.1%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 141 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
Whole-genome Sequencing	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
Shotgun Sequencing	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
Other Types	0.37	0.43	0.49	0.56	0.64	0.73	0.83	0.95	14.3%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 142 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.67	1.97	2.32	2.73	3.21	3.80	4.50	5.34	18.2%
Sequencing by Ligation	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
Nanopore Sequencing	0.19	0.23	0.29	0.35	0.43	0.53	0.65	0.80	22.8%
Other Technologies	1.37	1.60	1.86	2.17	2.53	2.96	3.47	4.08	17.0%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 143 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.17	1.39	1.66	1.97	2.35	2.80	3.36	4.03	19.5%
Academic and Research Institutes	1.68	1.97	2.30	2.69	3.15	3.70	4.35	5.14	17.4%
Other End Users	0.86	1.00	1.17	1.36	1.58	1.84	2.16	2.53	16.8%
Total	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.3 INDIA

10.4.3.1 Increasing prevalence of chronic diseases to support growth

The microbiome sequencing services market in India is experiencing significant growth due to several drivers. One of the major drivers is the increasing prevalence of chronic diseases in the country. According to the World Health Organization, in 2019, non-communicable diseases such as cardiovascular diseases, diabetes, and cancer accounted for 66% of all deaths in India. The microbiome has been shown to play a crucial role in the development and progression of these diseases, making microbiome sequencing services an important tool in disease prevention and management.

Another driver is the growing demand for personalized medicine. With advancements in microbiome research, there is a growing understanding of the unique microbiome profiles of individuals and how they can impact their health. Microbiome sequencing services can provide personalized insights into an individual's microbiome composition and help in developing personalized treatment plans.

Government initiatives and funding programs are also driving the growth of the microbiome sequencing services market in India. The Department of Biotechnology (DBT) has launched several programs to promote microbiome research in the country, such as the Human Microbiome Project and the Microbial

Culture Collection program. These initiatives have created opportunities for the development of new microbiome sequencing services and technologies in India.

One of the major restraints for market growth in India is the lack of awareness and understanding of the microbiome among the public and healthcare professionals. This is limiting the adoption of microbiome sequencing services and their incorporation into clinical practice.

TABLE 144 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Sequencing and Library Preparation	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Data Analysis	0.07	0.08	0.10	0.11	0.13	0.14	0.17	0.19	15.0%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 145 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Whole-genome Sequencing	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Shotgun Sequencing	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Other Types	0.05	0.05	0.06	0.07	0.08	0.09	0.10	0.11	13.2%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 146 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Sequencing by Ligation	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Nanopore Sequencing	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Other Technologies	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 147 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.15	0.18	0.21	0.25	0.29	0.35	0.41	0.49	18.7%
Academic and Research Institutes	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Other End Users	0.11	0.13	0.15	0.17	0.19	0.23	0.26	0.30	15.8%
Total	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.4 REST OF ASIA PACIFIC

The RoAPAC includes Australia, Taiwan, New Zealand, Thailand, South Korea, Singapore, and Malaysia, among other Asia Pacific countries. Factors such as public and private initiatives to support the adoption and development of microbiome technologies, increasing R&D expenditure, and increasing awareness of the diagnostic applications of microbiome sequencing are driving the growth of the RoAPAC market.

In September 2019, the South Korean government proposed plans to increase the budget of government subsidies in the biotech sector by 16% and finance a project to secure the genome data of 1 million people. The government aims to increase its budget spending to help the pharmaceutical sector develop innovative new drugs and medical devices. As per this agenda, the government will invest USD 951.8 million in R&D projects in the sector in 2020, with an increase of 16% in 2019. Over the past few years, a few events focusing on spreading awareness about the benefits of microbiome sequencing and the latest advancements in this field have been organized in the RoAPAC region.

In June 2022, Singapore-based precision gut microbiome company AMILI received USD 10.5 million in a Series A funding round led by technology and life sciences investor Vulcan Capital. New investors also joined the round, including Pruksha Group, TVM Capital Healthcare, Emtex Group, Capital Code, Pureland Group, Blue7, and Enterprise Singapore's investment arm SEEDS Capital.

TABLE 148 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Sequencing and Library Preparation	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Data Analysis	1.03	1.18	1.35	1.54	1.76	2.02	2.32	2.68	14.7%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 149 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Whole-genome Sequencing	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Shotgun Sequencing	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Other Types	0.68	0.77	0.87	0.97	1.10	1.23	1.39	1.57	12.7%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 150 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.01	3.51	4.10	4.78	5.58	6.53	7.67	9.03	17.1%
Sequencing by Ligation	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Nanopore Sequencing	0.34	0.42	0.51	0.63	0.76	0.93	1.14	1.40	22.2%
Other Technologies	2.48	2.86	3.30	3.81	4.40	5.09	5.90	6.88	15.8%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 151 **REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)**

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.10	2.49	2.94	3.48	4.11	4.88	5.80	6.92	18.7%
Academic and Research Institutes	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Other End Users	1.55	1.79	2.06	2.38	2.74	3.16	3.66	4.25	15.6%
Total	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.5 ASIA PACIFIC: IMPACT OF RECESSION

The impact of the recession on the global market would affect the Asia Pacific region as well. The Association of Southeast Asian Nations (ASEAN) projected a steady decline in the economic growth of Asia Pacific countries owing to the slower growth of their trading partners, namely, Europe, the US, and China. The overall GDP decline in the coming years is expected to affect the healthcare expenditure, spending, and wages in the industry with reduced import and export activities.

However, in Asia, countries focus on increasing their R&D in the pharmaceutical and biotechnology industry. India has seen a progression in terms of its R&D funding and rising government support. According to a report by India Brand Equity Foundation (IBEF), India's domestic pharmaceutical market was estimated at USD 42 billion in 2021. It is likely to reach USD 65 billion by 2024 and may further expand to reach USD 120–130 billion by 2030. Sun Pharma, a leading pharmaceutical company in India, announced that it intends to invest USD 600–650 million in R&D. On the other hand, Chinese pharmaceutical companies invested nearly 10% of their sales revenue in R&D in 2021.

For the overall Asia Pacific region, the impact of the recession is expected to be low during the forecast period. Japan's economy is expected to be more stable than other Asian countries. However, price inflation leading to lower consumption would lead to a slight decline in Japan's economic growth in 2023.

10.5 LATIN AMERICA

The Latin American microbiome sequencing services market is expected to reach USD 8.88 million by 2028 from USD 5.26 million in 2023, at a CAGR of 11.1% from 2023 to 2028.

The Latin American microbiome sequencing services market has been growing steadily in recent years. The region is witnessing an increased prevalence of chronic diseases, such as cancer, IBD, and diabetes, which have been linked to imbalances in the gut microbiome. As a result, there is a growing demand for microbiome sequencing services to better understand and manage these conditions. Additionally, the region's large and diverse population provides a unique opportunity for microbiome research and discovery.

The lack of awareness and understanding of the potential benefits of microbiome sequencing among healthcare providers and the general population is a key factor restraining the growth of this regional market. This has resulted in a relatively small market size and limited investment in the field. Additionally, the high cost of sequencing and the lack of standardization and regulation of microbiome testing can be a barrier to entry for smaller companies.

TABLE 152 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%
Rest of Latin America	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 153 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%
Sequencing and Library Preparation	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%
Data Analysis	0.66	0.72	0.78	0.85	0.92	1.00	1.09	1.19	8.7%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 154 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Whole-genome Sequencing	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Shotgun Sequencing	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Other Types	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 155 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.93	2.14	2.37	2.63	2.92	3.25	3.62	4.05	11.3%
Sequencing by Ligation	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%
Nanopore Sequencing	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%
Other Technologies	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 156 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.34	1.51	1.71	1.92	2.16	2.44	2.76	3.13	12.9%
Academic and Research Institutes	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%
Other End Users	1.00	1.09	1.19	1.31	1.43	1.56	1.72	1.89	9.6%
Total	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.1 BRAZIL

10.5.1.1 Growing adoption of next-generation sequencing technologies to propel growth

Brazil has been increasingly adopting NGS technologies for microbiome sequencing, which offer higher throughput, greater accuracy, and lower cost compared to traditional sequencing methods. This has led to the development of new microbiome sequencing services and the expansion of existing ones in the country.

The increasing prevalence of chronic diseases such as diabetes, obesity, and IBD is another major driver for market growth in Brazil, as it has created a high demand for microbiome sequencing services to better understand the role of the microbiome in disease development and progression.

TABLE 157 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Sequencing and Library Preparation	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Data Analysis	0.31	0.34	0.37	0.40	0.43	0.47	0.51	0.56	8.9%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 158 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Whole-genome Sequencing	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Shotgun Sequencing	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Other Types	0.21	0.22	0.24	0.25	0.27	0.29	0.31	0.33	6.9%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 159 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.91	1.01	1.12	1.24	1.38	1.53	1.71	1.92	11.4%
Sequencing by Ligation	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Nanopore Sequencing	0.10	0.12	0.14	0.16	0.19	0.22	0.26	0.30	16.5%
Other Technologies	0.75	0.82	0.90	0.99	1.08	1.19	1.31	1.45	10.0%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 160 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.63	0.71	0.80	0.90	1.02	1.15	1.30	1.47	12.9%
Academic and Research Institutes	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Other End Users	0.47	0.51	0.56	0.62	0.67	0.74	0.81	0.89	9.7%
Total	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.2 REST OF LATIN AMERICA

The Rest of Latin America mainly comprises Argentina, Colombia, Costa Rica, and Peru. The microbiome sequencing services market in the Rest of Latin America is estimated to reach USD 4.69 million by 2028 from USD 2.79 million in 2023, at a CAGR of 11.0% from 2023 to 2028.

Market growth in this region will be primarily driven by the awareness of and acceptance of personalized medicine, the increasing prevalence of chronic diseases (such as diabetes, cancer, and gastrointestinal disorders), and increasing investment in genomics research by governments and private organizations. However, certain regulatory barriers, such as complex approval processes, may create delays or difficulties in launching new microbiome sequencing services. Moreover, there may be a shortage of qualified professionals with the necessary expertise to perform and interpret microbiome sequencing.

TABLE 161 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Sequencing and Library Preparation	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Data Analysis	0.35	0.38	0.41	0.45	0.48	0.53	0.57	0.62	8.5%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 162 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Whole-genome Sequencing	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Shotgun Sequencing	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Other Types	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.36	6.4%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 163 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.02	1.13	1.26	1.39	1.54	1.71	1.91	2.13	11.2%
Sequencing by Ligation	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Nanopore Sequencing	0.12	0.14	0.16	0.19	0.22	0.25	0.29	0.34	16.6%
Other Technologies	0.84	0.93	1.01	1.11	1.21	1.33	1.46	1.61	9.8%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 164 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.71	0.80	0.90	1.02	1.14	1.29	1.46	1.66	12.9%
Academic and Research Institutes	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Other End Users	0.53	0.58	0.63	0.69	0.75	0.82	0.90	0.99	9.5%
Total	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.3 LATIN AMERICA: IMPACT OF RECESSION

A US recession would affect Latin America's real GDP outlook by impacting trade and the overall economy. GDP growth of 3.3% in 2022 is expected to slow by 1.5% in 2023 in Latin America. However, government bodies and pharmaceutical companies are expanding the pharmaceutical and biotechnology industry in countries such as Brazil. In October 2022, Luis Inácio Lula da Silva, one of the candidates in the presidential election and winner of Brazil's first-round elections, pledged to increase healthcare spending by investing in the Unified Health System (SUS) and health infrastructure, thereby expanding the domestic pharmaceutical industry. In November 2022, MedQuímica Indústria Farmacêutica (MedQuímica), a wholly owned subsidiary of Lupin in Brazil, signed a definitive agreement to acquire all rights to nine medicines from BL Indústria Ótica Ltda., a subsidiary of Bausch Health Companies Inc., to enhance its product portfolio in Brazil. Such initiatives are likely to encourage the growth of the pharmaceutical and biotechnology industry in 2023 and 2024.

Inflation has reached an inflection point in most countries in Latin America with tighter monetary conditions, stabilizing prices for energy, health and food commodities, transportation costs, and supply chain disruptions. All these factors would create an impact on the pharmaceutical and biotechnology industry and related technologies, such as metagenomic sequencing. Market growth is expected to slow in 2023-24 and will eventually start to stabilize and recover in 2025-26.

10.6 MIDDLE EAST & AFRICA

The microbiome sequencing services market in the Middle East & Africa is projected to reach USD 5.55 million by 2028 from USD 3.55 million in 2023, at a CAGR of 9.3% from 2023 to 2028.

The increasing awareness and understanding of the importance of the microbiome in human health and diseases are contributing to the growth of this regional market. However, the market growth is likely to be restrained by factors such as the high costs associated with microbiome sequencing services and a lack of skilled professionals in the region. Despite these challenges, there are several opportunities for market growth in MEA, including the potential for increased investment in research and development, partnerships and collaborations between academic and industrial sectors, and the development of novel technologies and analytical tools.

TABLE 165 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Sequencing and Library Preparation	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Data Analysis	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 166 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Whole-genome Sequencing	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Shotgun Sequencing	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Other Types	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis